

Virtual Mitosis Lab

Name _____

Find the virtual lab at:

http://www.biology.arizona.edu/cell_bio/activities/cell_cycle/cell_cycle.html

Answer the questions below as you read the first two pages of the virtual lab website.

1. Why are root tips so useful for observing mitosis? _____

Root tips are meristem tissue, so many cells are actively dividing. This high mitotic index makes mitosis easy to observe.

2a. Describe what happens to the DNA in interphase. _____

_____ 2b. Are the chromosomes visible? _____

3a. In what phase do the chromosomes become visible? _____

b. What happens to the nuclear membrane at this point? _____

4. During metaphase, why are the chromosomes organized to line up in the middle of the cell? _____

Chromosomes line up at the metaphase plate so each daughter cell receives one copy of every chromosome when chromatids separate.

5. What is the main characteristic of anaphase? _____

Sister chromatids separate and move to opposite poles of the cell.

6a. What are two specific events that occur in telophase? _____

Two events in telophase: (1) nuclear membranes reform around each set of chromosomes, and (2) chromosomes decondense. Cytokinesis also begins.

➤ Click NEXT at the bottom of the page. You do not need to copy the data table onto separate paper, as it's included here.

➤ Now you will begin the lab. You will be presented with 36 onion root tip cells. Determine which phase each cell is in. (You will know if you are correct or not.) Place a tally mark in the appropriate column in the table below. You should have a total of 36 tally marks when you are finished.

Interphase	Prophase	Metaphase	Anaphase	Telophase

➤ Complete the data table below by following the instructions for each row.

Data Table

	Interphase	Prophase	Metaphase	Anaphase	Telophase	Totals
# of cells	20	10	3	2	1	36
% of cells	55.56%	27.78%	8.33%	5.56%	2.78%	100%
# of minutes	588.9	294.4	88.3	58.9	29.4	1060

You'll need a calculator to complete some of the rows in this table.

- To fill in the “**# of cells**” row, count the number of cells in each of the phases from the data table on page 1 of this lab and write the number in the appropriate column.
- To fill in the “**% of cells**” row, divide the number of cells in each phase by 36 and then multiply by 100. (We use 36 because that's the total number of cells you observed.)
- To fill in the “**# of minutes**” row you'll determine how long a cell spends in each phase. To do this, multiply the % of cells by 1060, which is the total number of minutes in an onion root tip cell cycle.
- You can check your work for each row by adding up the numbers in each column. The total should be exact in the # of cells row, and extremely close, if not exact, in the other two rows.

Analysis Questions

1. Which phase of the cell cycle had the most cells? _____. Why do you think this is so? _____

Interphase had the most cells because cells spend the longest time growing, doing normal functions, and replicating DNA before division.

2. Describe how the cells in interphase looked compared to the cells in prophase.

Interphase cells had an intact nucleus with diffuse chromatin (no clear chromosome shapes). Prophase cells showed condensed, visible chromosomes and the nuclear membrane breaking down.

3. Describe how the cells in metaphase looked compared to the cells in anaphase.

In metaphase, chromosomes are lined up across the middle of the cell. In anaphase, sister chromatids split and move toward opposite poles.

4. Are the daughter cells produced by mitosis identical to their parent cell or are they different? Explain your answer.

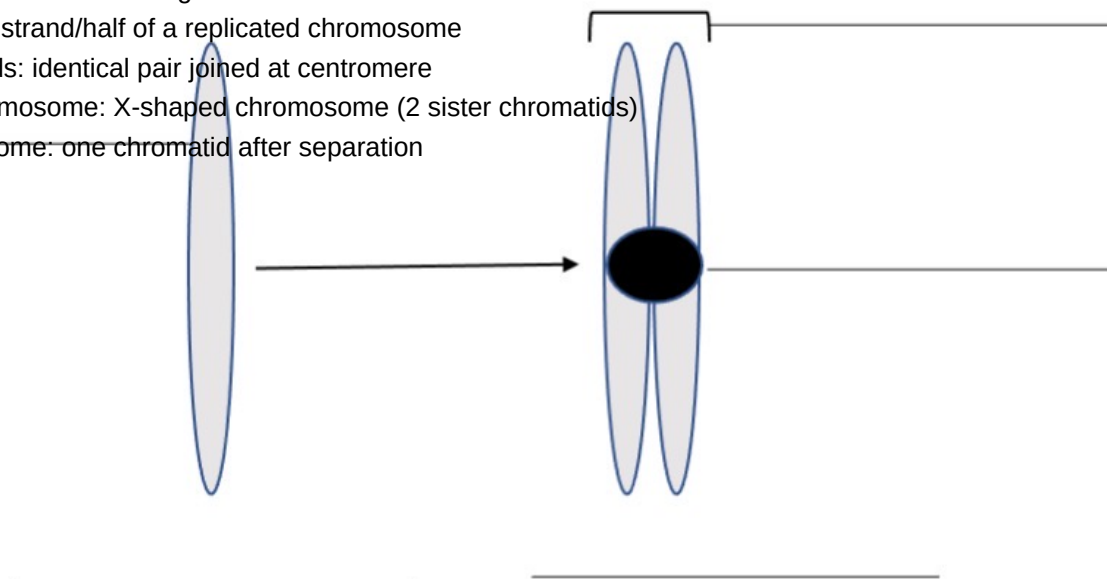
The daughter cells are genetically identical to the parent cell (except rare mutations) and keep the same chromosome number.

5. Label the illustration below using the following terms:

centromere, chromatid, sister chromatids, replicated chromosome, single chromosome

Label key:

- Centromere: pinched middle region where chromatids are attached
- Chromatid: one strand/half of a replicated chromosome
- Sister chromatids: identical pair joined at centromere
- Replicated chromosome: X-shaped chromosome (2 sister chromatids)
- Single chromosome: one chromatid after separation



6. What process causes the single chromosome to copy itself? _____
DNA replication during S phase of interphase.

7. How many times do the chromosomes replicate in mitosis? _____
Once.

8. How many daughter cells are produced during mitosis? _____
Two daughter cells.

9. What is the term used to describe the "pinching in" of the cell membrane during telophase?
Cytokinesis (formation of a cleavage furrow in animal cells).

10. In what phase do the chromosomes replicate? _____
Interphase (specifically S phase).

Label the phase each cell represents. Then describe the characteristics of each cell that helped you decide.

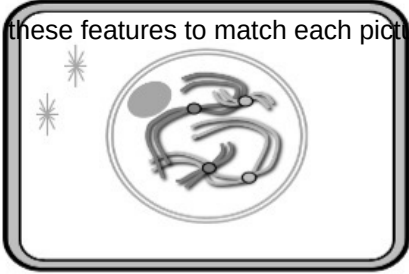
A. _____

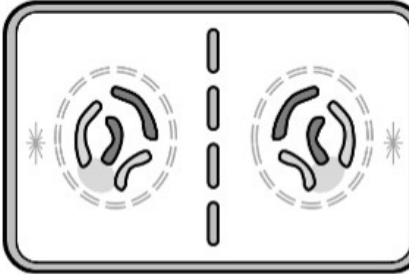
Suggested phase labels for typical mitosis cells:

- 1) Interphase - intact nucleus, diffuse chromatin, no lined-up chromosomes
- 2) Prophase - chromosomes condense and become visible; nuclear membrane breaks down
- 3) Metaphase - chromosomes lined up across the center (metaphase plate)
- 4) Anaphase - sister chromatids separate and move to opposite poles
- 5) Telophase - two nuclei reform; chromosomes decondense; cytokinesis begins

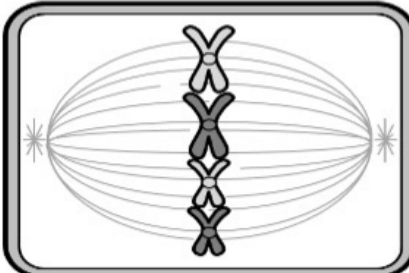
B. _____

Use these features to match each pictured cell to its phase.





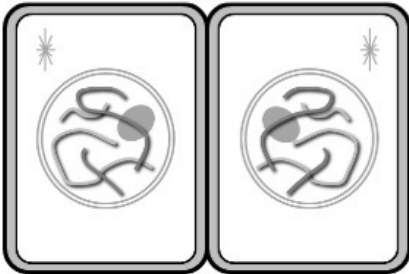
C. _____



D. _____



E. _____



F. _____
